retracted his objection. Nevertheless, it is clear that Einstein did not think Friedmann's results had any relevance to the universe. But about five years later, Hubble's detailed observations of a few dozen galaxies with the hundred-inch telescope at Mount Wilson Observatory confirmed that, indeed, the universe is expanding. Friedmann's work, refashioned in a more systematic and efficient form by the physicists Howard Robertson and Arthur Walker, still forms the foundation of modern cosmology.

In a little more detail, the modern theory of cosmic origins goes like this. Some 15 billion or so years ago, the universe erupted from an enormously energetic, singular event, which spewed forth all of space and all of matter. (You don't have to search far to locate where the big bang occurred, for it took place where you are now as well as everywhere else; in the beginning, all locations we now see as separate were the same location.) The temperature of the universe a mere 10-43 seconds after the bang, the so-called Planck time, is calculated to have been about 1032 Kelvin, some 10 trillion trillion times hotter than the deep interior of the sun. As time passed, the universe expanded and cooled, and as it did, the initial homogeneous, roiling hot, primordial cosmic plasma began to form eddies and clumps. At about a hundred-thousandth of a second after the bang, things had cooled sufficiently (to about 10 trillion Kelvin—about a million times hotter than the sun's interior) for quarks to clump together in groups of three, forming protons and neutrons. About a hundredth of a second later, conditions were right for the nuclei of some of the lightest elements in the periodic table to start congealing out of the cooling plasma of particles. For the next three minutes, as the simmering universe cooled to about a billion degrees, the predominant nuclei that emerged were those of hydrogen and helium, along with trace amounts of deuterium ("heavy" hydrogen) and lithium. This is known as the period of primordial nucleosynthesis.

Not a whole lot happened for the next few hundred thousand years, other than further expansion and cooling. But then, when the temperature had dropped to a few thousand degrees, wildly streaming electrons slowed down to the point where atomic nuclei, mostly hydrogen and helium, could capture them, forming the first electrically neutral atoms. This was a pivotal moment: from this point forward the